Yokesh Ramasamy Bhalamuruhan

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Portfolio: yrb.web.app | LinkedIn: linkedin.com/in/yokeshrb | GitHub: github.com/Yokesh-R-B

Summary

Software Developer with **3+ years of combined experience** across software development, RPA, data engineering, and workflow automation. Experienced in building Python-based backend applications, enterprise data migration tools, API integrations, distributed workflows, and AI-powered automation. Skilled in Python, Java, SQL, REST APIs, and workflow orchestration, with experience at BMW Group, FORVIA HELLA, and Tata Consultancy Services.

Technical Skills

Programming Languages: Python, Java, SQL, TypeScript
Backend & APIs: FastAPI, Spring Boot, REST APIs, GraphQL, Hibernate, JPA
Data Engineering: ETL, Data Pipelines, Data Migration, Data Validation, Batch Processing, Pandas, NumPy, PostgreSQL, MySQL, MongoDB
AI & Automation: LLMs, Retrieval-Augmented Generation (RAG), LangChain, n8n, Workflow Automation
Testing & Infrastructure: Selenium WebDriver, Docker, Linux, Git, Maven, CI/CD, Azure
Engineering Concepts: Distributed Systems, Microservices, System Design, OOP, Data Structures & Algorithms
Digital Engineering: NVIDIA Omniverse, Blender

Professional Experience

BMW Group, Munich Germany
Artificial Intelligence Developer – Virtual Factory *09/2026 – Present*

- Developed Python-based **data migration health checks** for factory building and layout data migration from ProjectWise to BMW's centralized factory-data platform (SW8P), validating **80 plants and 325K+ files** through REST APIs, OAuth/M2M authentication, automated validation, and Excel reporting
- Built a Python-based **bulk user and project-member migration tool** using GraphQL APIs, implementing dry-run validation, audit logging, and idempotent execution to prevent duplicate assignments
- Developed production-ready **n8n workflow automations** integrating Jira, BMW LLM APIs, and Microsoft Outlook to detect unresolved defect discussions, prioritize follow-ups, automate reminders, and prevent duplicate notifications
- Built a **Python/FastAPI and Leaflet web application** integrating the NavVis IVION API to visualize 3D scan coverage, scan frequency, last-scanned dates, and unscanned areas across BMW plants, buildings, and floors
- Worked on AI-assisted factory layout planning workflows using **Blender and GPT Astra** for 3D model generation and layout optimization

FORVIA Hella GmbH & Co. KGaA, Lippstadt Germany
Software Developer / Master's Thesis *10/2024 – 09/2025*

- Designed and implemented a distributed **Python workflow engine** orchestrating automotive headlamp simulation jobs across HPC clusters, with modular **job scheduling, monitoring, result aggregation, and post-processing**
- Integrated the workflow engine with an **SPDM (Simulation Process and Data Management)** system, connecting simulation execution, engineering data, and traceability through REST API integrations and data pipelines
- Automated end-to-end simulation workflows, reducing manual processing effort by approximately **90%**

Independent Freelance Projects Germany
Freelance Automation Developer – n8n *10/2025 – 08/2026*

- Designed and developed automated business workflows using **n8n**, integrating APIs and business applications to streamline repetitive processes

- Implemented workflow logic involving data transformation, conditional routing, automated notifications, and error handling to improve process reliability
- Developed **supply chain automation workflows** to process incoming purchase order files, extract and validate order information, and automate order notifications and status updates through API and email integrations

Bauhaus-Universität Weimar Germany
Student Assistant *04/2024 – 09/2024*

- Enhanced a **Retrieval-Augmented Generation (RAG)** system by integrating external knowledge sources and improving information retrieval
- Participated in the **TREC 2024 RAG Track**, working on RAG evaluation, information retrieval, and optimization

Tata Consultancy Services, Chennai India
Assistant System Engineer – RPA Development (Java & Selenium WebDriver) *04/2021 – 02/2023*

- Developed and maintained **Java-based automation solutions using Selenium WebDriver** for a US-based financial services client, automating web-based workflows and validating high-volume transaction processing (approximately 100,000 transactions per day)
- Automated **financial transaction reconciliation and validation** against external records, identifying discrepancies and reducing manual testing and exception-handling effort by **40%**
- Automated and validated **end-of-day settlement and batch-processing workflows**, improving the accuracy and reliability of financial data processing

Education

Bauhaus-Universität Weimar Germany
Master of Science in Digital Engineering (Current Grade: 1.6) *04/2023 – 11/2026 (Expected)*

- Master's thesis completed 09/2025 at FORVIA Hella; final presentation pending
- Coursework: Machine Learning, Computer Vision, Simulation Methods, Stochastic Systems, Optimization, Software Engineering

Anna University Chennai India
Bachelor of Engineering in Civil Engineering; co-author, Int. Journal of Steel Structures (2024)04/2017 – 03/2021

Projects

Personal Portfolio Website | *yrb.web.app*

- Developed a personal portfolio website presenting software projects, technical implementations, and professional experience

RAG Chatbot Backend Application | *Python, FastAPI, LangChain, ChromaDB, SQLite, LLM*

- Built an end-to-end **RAG pipeline** with FastAPI endpoints, ChromaDB vector search, and SQLite persistence for multi-turn chat

Languages

English (C1), German (B1), Tamil (Native)